Tutorial 1: Steady Laminar Pipe Flow

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CLL769: Applications of CFD

1 Problem Statement & Objective

The objective is to perform numerical simulations of a steady, laminar flow of a Newtonian, incompressible fluid in a pipe. The system involves a pipe of length 1 m and radius 0.1 m operating in the laminar regime at $Re = 100$. An axisymmetric 2D simulation is performed to validate CFD predictions against analytical solutions (Hagen-Poiseuille flow) and analyze wall shear stress.

2 Governing Equations

The flow is governed by the steady-state, incompressible Navier-Stokes equations for a Newtonian fluid:

$$\nabla \cdot \vec{U} = 0 \quad \text{(Continuity)} \tag{1}$$

$$\rho(\vec{U} \cdot \nabla)\vec{U} = -\nabla P + \mu \nabla^2 \vec{U} \quad \text{(Momentum)} \tag{2}$$

In the axisymmetric cylindrical coordinate system, the azimuthal velocity and θ-derivatives are zero, reducing the equations to 2D (r, z).

3 Computational Setup

The boundary conditions applied are a velocity inlet and a pressure outlet (0 gauge). The top edge is a wall with a no-slip condition, and the bottom edge represents the axis of symmetry. To achieve $Re = 100$ with $D = 0.2$ m, fluid properties were set to $\rho = 1$ kg/m^3, $U = 0.5$ m/s, and $\mu = 0.001$ kg/m-s.

A pressure-based, steady, axisymmetric solver was used with the SIMPLE algorithm for pressure-velocity coupling and the QUICK scheme for spatial discretization of momentum.

4 Geometry & Meshing

An axisymmetric 2D structured mesh was created for the pipe. A bias factor was applied towards the wall to adequately resolve the velocity gradients in the boundary layer.

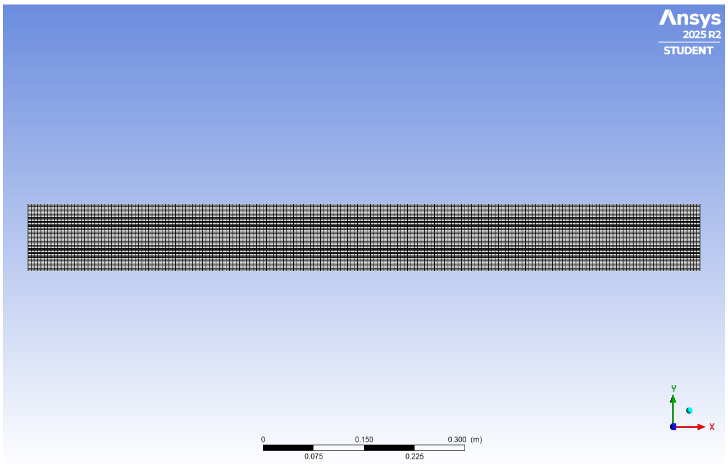

Figure 1: Representative view of the computational grid.

5 Results & Discussion

5.1 Velocity Distribution

The flow enters with a uniform velocity and gradually develops into a parabolic profile. The maximum velocity occurs at the axis, and zero velocity at the wall due to the no-slip condition.

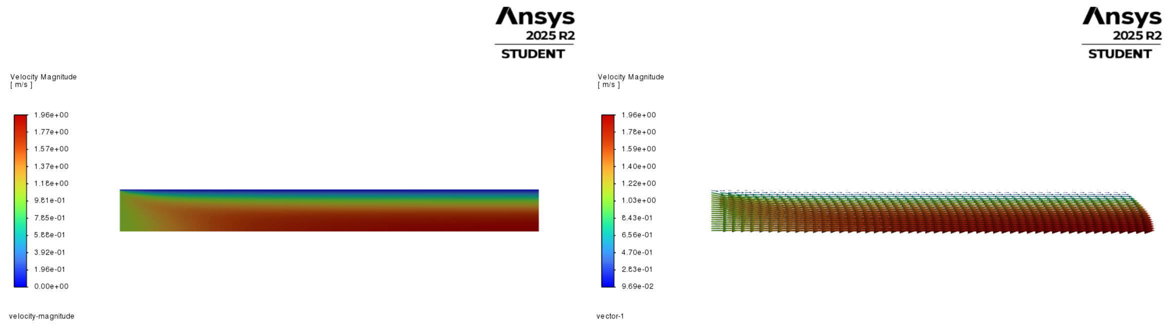

Figure 2: Velocity magnitude contour (left) and vectors (right).

5.2 Validation: Velocity Profile & Pressure Drop

The predicted fully developed profile matches the analytical parabolic solution perfectly:

$$u(r) = 2(0.5)\left(1 - \left(\frac{r}{0.1}\right)^2\right) = 1.0\left(1 - 100r^2\right)\ \text{m/s}$$

For $U_{\text{avg}} = 0.5$ m/s, the analytical maximum velocity at the axis ($r = 0$) is $u_{max} = 1.0$ m/s, which perfectly matched the CFD predictions. Furthermore, using the Hagen-Poiseuille equation:

$$\frac{\Delta P}{L} = \frac{8\mu U_{\text{avg}}}{R^2} = \frac{8(0.001)(0.5)}{(0.1)^2} = 0.4\ \text{Pa/m}$$

The simulated pressure drop exactly matched this calculated analytical value of 0.4 Pa/m in the fully developed region.

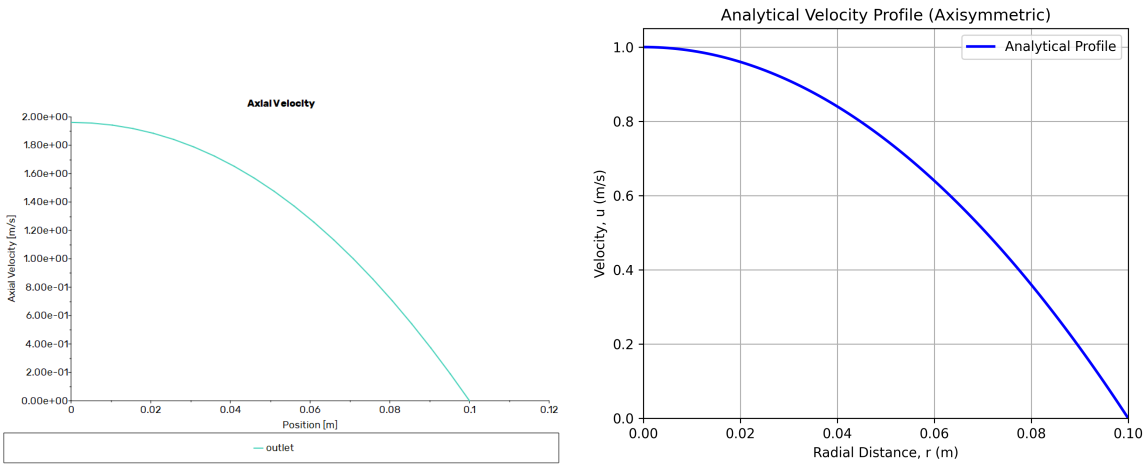

Figure 3: Fully developed radial velocity profile. Left: Simulated. Right: Analytical.

5.3 Wall Shear Stress

Near the inlet, the velocity gradient at the wall is extremely steep, resulting in high wall shear stress. As the boundary layer grows and the flow develops, the wall shear stress decreases exponentially, eventually reaching a constant horizontal asymptote once the flow is fully developed.

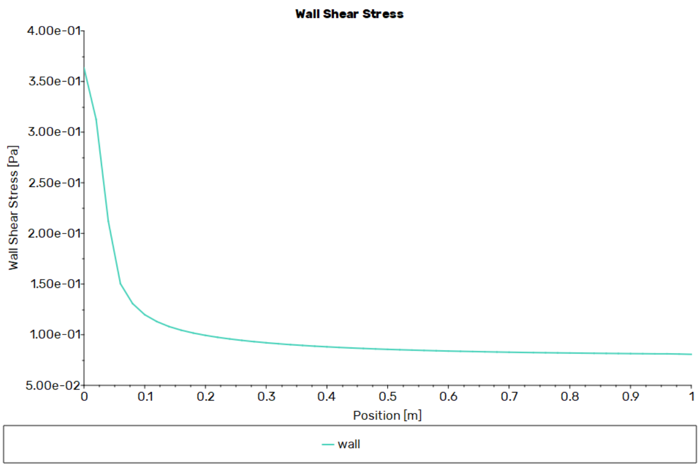

Figure 4: Variation of wall shear stress along the pipe length.